

PROJECT CONTROLS INSTITUTE

PCP-AI — Candidate Information Pack

Everything a candidate needs to understand and prepare for the Certified Project Controls Professional — AI examination: the content outline, the candidate handbook, an overview of the credential, and sample questions.

Version 1 · for publication on PCI platforms
Fees, dates and the final examination specification are confirmed on the official PCI website.

PCP-AI Examination — Content Outline (Blueprint)

Certified Project Controls Professional — AI (PCP-AI) · Project Controls Institute

Version 1 — published for candidates and training providers. This outline is the authoritative statement of what the PCP-AI examination assesses. It is drawn from, and maps one-to-one to, the PCP-AI Body of Knowledge. Weightings shown are indicative for exam-form assembly and are confirmed by the Institute for each published form.

1. Purpose

The PCP-AI credential certifies that a project-controls professional can apply sound financial-reporting and accounting judgement, the full project-management and controls toolkit (budgeting, earned value, scheduling, contracts, risk and process controls), and the **governed use of artificial intelligence** in project controls. The governing principle across the AI content is simple and non-negotiable: **AI proposes; the professional disposes** — the human professional remains accountable for every estimate, forecast, commitment and decision.

2. Examination at a glance

Item	Detail
Questions	Multiple-choice, four options, one correct answer
Duration	90 minutes
Pass mark	65%
Delivery	Online, remotely proctored (browser or the PCI Secure Exam desktop application)
Languages	Interface available in English, Arabic, Spanish, French, Korean, Chinese and Russian
Result	Issued immediately on submission, with a domain-by-domain performance breakdown
Scheduling window	12 months from payment of the examination fee

3. Competency areas & weighting

The examination is built on three competency areas. Their weighting reflects the balance a modern project-controls professional needs: rigorous financial grounding, deep project-management and controls capability, and the responsible application of AI.

Area	Weighting
A. Financial Reporting & Accounting	40%
B. Project Management, Controls & Delivery	40%
C. AI for Project Controls	20%

4. Domain & Knowledge-Area outline

Thirteen domains contain sixty-one Knowledge Areas. Indicative domain weightings are shown; every Knowledge Area listed is examinable.

Area A. Financial Reporting & Accounting — 40%

Domain 1 — Foundations of Accounting for Project Controls · ~8%

KA	Knowledge Area	What it covers
1.1	The accounting model	The accounting equation; debit & credit rules by account type; double-entry mechanics; the ledger and trial balance; worked postings through a full...
1.2	Components of the financial statements	Statement of Financial Position, Statement of Profit or Loss & OCI, Statement of Cash Flows, Statement of Changes in Equity, notes; what each shows...
1.3	Accrual accounting & the matching concept	Accruals vs cash basis; the matching principle; revenue/expense recognition timing; accrue/defer worked entries.
1.4	Cost provisions & cost accruals	Recognising provisions vs accruals; estimation; reversal; link to IAS 37; a worked warranty / contract-loss provision.
1.5	Chart of accounts & cost coding for projects	Cost breakdown structure, control accounts, mapping to WBS; project cost coding. AI in this domain: automated coding/classification, anomaly detect...

Domain 2 — Financial Reporting & the Standards · ~12%

KA	Knowledge Area	What it covers
2.1	The reporting framework	Purpose of financial reporting; qualitative characteristics; IFRS vs local GAAP overview; the role of IAS 1 (presentation).
2.2	IFRS 15 Revenue from Contracts with Customers — in depth	FLAGSHIP KA. The five-step model (identify contract; identify performance obligations; determine transaction price; allocate; recognise as/when sat...
2.3	Revenue recognition beyond IFRS 15 basics	Principal vs agent; bundled goods/services; link to billing and BoQ (cross-ref Domain 7).
2.4	Other relevant standards for project controls	IAS 37 (provisions, contingent liabilities/assets), IAS 2 (inventories/WIP where relevant), IAS 16 / IFRS 16 (assets/leases as they touch project c...
2.5	Management reporting vs statutory reporting	Internal vs external; management accounts. AI in this domain: drafting disclosures, consistency checks, standard-mapping assistants (with human sig...

Domain 5 — Cost Management & Cost Control · ~8%

KA	Knowledge Area	What it covers
5.1	Cost management framework	Cost types (direct/indirect, fixed/variable), cost drivers, allocation & absorption; overhead recovery.
5.2	Cost control cycle	Commitment -> accrual -> actual; cost ledger; cost extraction and reconciliation from the ERP/source systems; data integrity.
5.3	Cost breakdown & control accounts	CBS, control accounts, work packages (cross-ref WBS).
5.4	Change control & cost impact	Trends, variations, change orders and their cost effect. AI in this domain: automated cost extraction, reconciliation, outlier detection.

Domain 7 — Contracts, Commercial Management, BoQ, Invoicing & Revenue · ~8%

KA	Knowledge Area	What it covers
7.1	Types of contract	Lump sum/fixed price, remeasurement, cost-plus (CPFF/CPIF/CPAF), time & materials, unit-rate, target-cost/pain-gain, framework/EPC; risk allocation...
7.2	Contract management	The contract lifecycle; obligations, milestones, claims, liquidated damages, retention, bonds/guarantees; FIDIC awareness (concept level, no verbat...
7.3	Bills of Quantities (BoQ)	Structure; measurement; rates; preliminaries; pricing a BoQ; worked: price a small BoQ and a remeasurement example.
7.4	Invoicing & applications for payment	Interim valuations, progress billing, certification, retention release; linking BoQ/EVM progress to billing; worked: an interim payment application.
7.5	Revenue recognition in the commercial cycle	Tying billing to IFRS 15 (cross-ref 2.2); over-/under-billing (contract asset/liability); worked: over-billing vs revenue earned. AI in this domain...

Domain 11 — Business Process Cycles (O2C, P2P & the Control Environment) · ~4%

KA	Knowledge Area	What it covers
11.1	Order-to-Cash (O2C)	The cycle end-to-end; controls; link to revenue & receivables.
11.2	Procure-to-Pay (P2P)	The cycle end-to-end; controls; link to cost, commitments & payables.
11.3	Internal control & segregation of duties	Internal control & segregation of duties in these cycles; the audit trail. AI in this domain: process mining, control-exception detection, invoice/...

Area B. Project Management, Controls & Delivery — 40%**Domain 3 — Budgeting & Forecasting · ~7%**

KA	Knowledge Area	What it covers
3.1	Budgeting fundamentals	Purpose; top-down vs bottom-up; zero-based; the project budget baseline; contingency & management reserve.
3.2	Cost estimation	Estimate classes (order-of-magnitude to definitive; AACE class 1-5); methods (analogous, parametric, bottom-up); accuracy ranges; basis of estimate.
3.3	The time-phased budget / cost baseline (BCWS/PV)	Spreading cost over the schedule; the S-curve; worked: build a time-phased budget.
3.4	Forecasting	Forecasting methods; rolling forecasts; trend analysis; the Estimate at Completion (EAC) family and forecasting logic (deep EAC treatment cross-ref...
3.5	Cash flow forecasting	Project cash flow; inflows/outflows; funding. AI in this domain: ML-based cost/return forecasting, scenario generation, driver-based forecasting an...

Domain 4 — Performance Management, Variance Analysis & Management Reporting · ~6%

KA	Knowledge Area	What it covers
4.1	Performance management principles	KPIs, targets, thresholds, tolerances; leading vs lagging indicators.
4.2	Variance analysis	Cost/price/quantity/efficiency variances; favourable vs adverse; bridge analysis; worked: a full variance bridge from budget to actual.
4.3	Management reporting	Report design; the project controls dashboard; narrative vs numbers; reporting cadence; audience-appropriate reporting; exception reporting.
4.4	Data visualisation & storytelling for controls	Charts that inform decisions; common distortions to avoid. AI in this domain: automated commentary, exception detection, natural-language querying...

Domain 6 — Earned Value Management & Forecasting (EVM/EAC) · ~8%

KA	Knowledge Area	What it covers
6.1	EVM fundamentals	PV/BCWS, EV/BCWP, AC/ACWP; measurement methods (0/100, 50/50, % complete, milestones); worked: compute EV under each method.
6.2	Variances & indices	CV, SV, CPI, SPI (and TCPI); interpretation; fully worked.
6.3	Forecasting with EVM — the EAC family	ETC and EAC formulae ($EAC = AC + (BAC - EV)$; $EAC = BAC / CPI$; $EAC = AC + (BAC - EV) / (CPI \times SPI)$; etc.), when to use each, VAC; assumptions & limits; mult...
6.4	Integrating cost & schedule	Linking EVM to the schedule; limitations of EVM; earned schedule concept (brief). AI in this domain: predictive EAC, early-warning models, driver a...

Domain 8 — Project Management Lifecycle · ~6%

KA	Knowledge Area	What it covers
8.1	Initiating	Business case, charter, stakeholder identification, feasibility, success criteria.
8.2	Planning	Scope & WBS, schedule, cost/budget, quality, resource, risk, procurement, communications, integration; the baselines.
8.3	Executing	Directing work, managing resources/stakeholders, quality assurance, procurement execution.
8.4	Monitoring & Controlling	Integrated change control, performance measurement (ties to EVM/variance), controlling scope/schedule/cost/quality/risk.
8.5	Closing	Contract & project closure, handover, lessons learned, final account; a threaded mini-case runs a single project through all five process groups.
8.6	Development approaches — predictive, iterative, incremental & adaptive	The spectrum from waterfall/predictive to adaptive/agile; incremental delivery (build and deliver in usable slices) vs iterative delivery (refine t...

Domain 9 — Agile, Scrum & Adaptive Delivery for Project Controls · ~5%

KA	Knowledge Area	What it covers
9.1	Agile foundations	The Agile Manifesto (values & 12 principles, described not reproduced); the agile mindset; empirical process control (transparency, inspection, ada...
9.2	The Scrum framework — in depth	Scrum theory & pillars; the three accountabilities (Product Owner, Scrum Master, Developers); the five events (the Sprint; Sprint Planning; Daily S...
9.3	Backlogs, estimation & agile metrics	User stories & acceptance criteria; INVEST; backlog refinement & prioritisation (MoSCoW, WSJF concept); relative estimation (story points, planning...
9.4	Kanban, Lean & scaling	Kanban (visualise, limit WIP, manage flow) and flow metrics; Lean principles & waste; scaling frameworks at awareness level (SAFe, LeSS, Scrum-of-S...
9.5	Agile cost control, forecasting & earned value in adaptive delivery	THE PROJECT-CONTROLS CRUX (cross-ref Domains 3 & 6). Funding models (fixed capacity/rolling-wave budgets vs fixed-scope); cost per sprint / run-rat...
9.6	Hybrid delivery & agile governance	Combining stage-gate/predictive governance with agile execution; milestone & phase-gate reporting over sprints; contracting for agile (T&M, capped...

Domain 10 — Project Scheduling · ~4%

KA	Knowledge Area	What it covers
10.1	Schedule development	Activity definition, sequencing, dependencies (FS/SS/FF/SF, leads/lags), estimating durations.
10.2	Network analysis & the Critical Path Method	Forward/backward pass, total & free float, critical path; fully worked CPM example with a network diagram described.
10.3	Schedule compression & resource	Crashing, fast-tracking; resource levelling/smoothing; schedule risk (PERT/Monte Carlo concept).
10.4	Progress measurement & schedule control	Updating, schedule variance, baselines vs actual; relating classical scheduling to agile cadence (sprints/releases as schedule increments - cross-r...

Domain 12 — Risk Management for Project Controls · ~4%

KA	Knowledge Area	What it covers
12.1	Risk framework	ISO 31000 principles; risk vs uncertainty; risk appetite/tolerance.
12.2	Risk process	Identification, qualitative & quantitative analysis (probability/impact, expected value, Monte Carlo concept), response planning, monitoring; the r...
12.3	Contingency & management reserve	Linking risk to budget (cross-ref 3.1). AI in this domain: risk identification from data, predictive risk scoring, simulation assistance.

Area C. AI for Project Controls — 20%

Domain 13 — AI for Project Controls & Project Management: Concepts, Tools & Practice · ~20%

KA	Knowledge Area	What it covers
13.1	AI foundations for professionals	What AI/ML/GenAI are; supervised/unsupervised/reinforcement learning at a working level; models vs rules; large language models (how they generate,...
13.2	Data — the fuel	Data quality, structure, governance, lineage; why 'garbage in, garbage out' dominates AI outcomes; project-controls data sources (ERP, schedule, co...
13.3	Prompting & working with GenAI	Effective prompting for professional tasks; structured prompts; giving context/examples; iterative refinement; verifying outputs; prompt patterns f...
13.4	AI tool categories for project controls & PM	Practical, tool-aware, vendor-neutral. Cover each category with what it does, representative tools, typical inputs/outputs, and governance notes: g...
13.5	AI applied across the project-controls lifecycle	The heart of the domain - cross-refs the finance & PM domains, going deeper than the per-chapter 'AI in this domain' boxes - hands-on workflows: es...
13.6	Governance, ethics, risk & assurance of AI	'AI proposes, the professional disposes' in depth: human accountability & sign-off; auditability (keeping the trail of what AI produced and who app...
13.7	Building an AI-augmented project-controls capability	Maturity model (from ad-hoc to governed); integrating AI into the controls workflow and tooling; upskilling; measuring value; pitfalls and change m...

5. Cognitive levels

Items are written across three cognitive levels: **recall** (definitions, principles, standards), **application** (compute a figure, apply a method to a scenario), and **analysis** (interpret results, choose between options, judge an AI output). Quantitative areas — earned value, variance analysis, forecasting, agile metrics, risk expected value — include numerical items in which the candidate computes an answer before selecting it.

6. How to prepare

Candidates study against the **PCP-AI Body of Knowledge**, which develops every Knowledge Area to full handbook depth with worked examples, figures, sample questions and self-check exercises. Sample questions published by the Institute are drawn from the same blueprint as, but kept separate from, the live examination bank, and are never reused verbatim as live items.

Standards are referenced by name at awareness level throughout the Body of Knowledge (IFRS, IAS, AACE Total Cost Management, PMBOK, ISO 31000, the Agile Manifesto, the Scrum Guide, COSO, the NIST AI Risk Management Framework and others). The Institute confirms the final blueprint, weightings and pass standard for each published examination form.

PCP-AI Candidate Handbook

Certified Project Controls Professional — AI (PCP-AI) · Project Controls Institute

Version 1 — published for candidates. This handbook explains what the PCP-AI credential is, how the examination works, and the policies that govern it. Figures such as fees and continuing-professional-development requirements are the Institute's current settings and are confirmed on the official PCI website at the time of registration.

1. About the credential

The **PCP-AI** certifies that a project-controls professional can combine three things that modern projects demand together: sound **financial-reporting and accounting** judgement, the full **project-management and controls** toolkit (budgeting, earned value, scheduling, contracts, risk and process controls), and the **governed use of artificial intelligence** in project controls.

The AI dimension is what sets the credential apart, and it is taught and examined honestly. The governing principle is "**AI proposes; the professional disposes**": AI can draft, summarise, detect and forecast, but the certified professional remains accountable for every estimate, commitment, forecast and decision, and is expected to know where AI helps, where it fails, and how to govern it.

2. Who it is for

The credential suits cost engineers, planners and schedulers, project and commercial controllers, quantity surveyors, PMO analysts, and finance professionals working on or alongside projects who want a recognised, current qualification that reflects both classical project controls and the responsible adoption of AI. There are no mandatory prerequisites to sit the examination; candidates should judge their readiness against the published Content Outline and Body of Knowledge.

3. Registration and fees

1. Create a PCI account and complete your profile.
2. Pay the **examination fee** — currently **USD 350** (confirmed at checkout; local pricing and any discounts are shown on the website). Payment opens a **12-month window** in which to schedule and sit the examination.
3. Provide a valid **government-issued photo ID** whose name matches your registration; identity is verified before the examination.

The examination fee covers one sitting. A further sitting (for example, after an unsuccessful attempt) requires a new examination fee.

4. The examination

Item	Detail
Format	Multiple-choice questions, four options, one correct answer
Duration	90 minutes
Pass mark	65%
Areas	A. Financial Reporting & Accounting (40%) · B. Project Management, Controls & Delivery (40%) · C. AI for Project Controls (20%)
Delivery	Online, remotely proctored — in the browser or via the PCI Secure Exam desktop application
Interface languages	English, Arabic (right-to-left), Spanish, French, Korean, Chinese and Russian
Aids	An on-screen calculator is provided; no other materials are permitted
Result	Issued immediately on submission, with a domain-by-domain performance breakdown

The examination is built from the published Content Outline. Questions span three cognitive levels — recall, application and analysis — and quantitative areas include numerical items that you compute before selecting an answer.

5. Booking, rescheduling and the window

- **Booking.** Choose a date and time inside your 12-month window. Launch opens **15 minutes before** your slot and stays open for a **30-minute grace period**.
- **Rescheduling.** You may reschedule **free of charge up to 72 hours before** your slot, up to **three times**. Bookings **lock 24 hours before** the start time.
- **The window cannot be extended.** If the 12-month window lapses without a sitting, a new examination fee is required.
- **Missed sittings.** If you do not launch within the grace period the booking is marked missed; documented incidents (for example, a genuine technical failure) are considered under the examination administration policy — open a support ticket with evidence.

6. On the day — readiness, identity and conduct

Before you launch you will complete a short **system readiness check** (camera, microphone, connection and screen) and an **identity check** against your photo ID. Access to your camera and microphone is requested only to confirm they work and to support proctoring; the readiness check itself records nothing.

During the examination:

- Work alone, in a quiet room, on a single screen, with notes and devices put away.
- The secure environment runs in full screen and monitors for leaving the examination window; copy, cut and paste are disabled. Losing focus or leaving the window is recorded as an integrity event on your attempt.
- The examination **cannot be quit** — submit when you are ready, or it submits automatically when time expires. Answers are saved continuously, so a brief disconnection does not lose your work or

reset the clock; you resume where you left off.

7. Results, scoring and the credential

- **Immediate result.** On submission you receive your score, a **pass/fail** outcome against the 65% standard, and a **domain-by-domain breakdown** showing where you performed above, at, or below target.
- **Immediate publication is the default.** Proctoring evidence is retained for audit but does not, by itself, delay your result. A result is only held back for a genuine **technical-validity** issue (for example, a late or invalid submission), and you are told plainly that no pass/fail outcome is shown until it is resolved.
- **The credential.** A pass issues a **PCP-AI credential** with a unique credential ID, verifiable online by anyone with the ID. Certificates and score reports are downloadable from your account.

8. Keeping the credential — recertification and CPD

The PCP-AI credential is valid for a **three-year cycle**. To keep it current you record **continuing professional development (CPD)** and recertify within the cycle. The current CPD target is set by the Institute (presently **30 hours per cycle**) and is shown, with your running total, in your account. Letting a credential lapse suspends its "active" status until recertification requirements are met.

9. Accommodations

The Institute supports candidates who need reasonable adjustments (for example, **extra time**, a **separate setting**, or **assistive technology**). Request an accommodation from your account **before booking**, describing what you need; requests are treated confidentially and reviewed by the Institute, and any approved extra time is applied to your examination automatically.

10. Integrity, appeals and data

- **Integrity.** The examination is proctored and the item bank is protected. Attempts to gain an unfair advantage, to capture or share examination content, or to have another person sit the examination are serious breaches and may invalidate a result and the credential.
- **Appeals and casework.** You may raise an appeal or a support case from your account; the Institute operates a documented casework process for accommodations, appeals and incidents.
- **Data and privacy.** Personal data, identity documents and proctoring evidence are handled under the Institute's privacy policy and retained only as long as needed for certification and audit. See the privacy policy on the website for detail.

Sample questions published by the Institute are drawn from the same blueprint as, but kept separate from, the live examination bank, and are never reused verbatim as live items. This handbook is a candidate-facing summary; where it and the Institute's formal regulations differ, the formal regulations on the official PCI website govern.

The PCP-AI Credential — Overview

Certified Project Controls Professional — AI · Project Controls Institute

A concise overview for prospective candidates, employers and training partners.

Project controls has changed. The credential should too.

Projects are still won or lost on the fundamentals — accurate cost, honest forecasts, a schedule you can trust, contracts that hold, risk that is understood. What has changed is that a new tool now sits on every controls professional's desk: **artificial intelligence**. Used well, it drafts reports, spots anomalies in a cost ledger, and sharpens a forecast. Used carelessly, it invents numbers with total confidence.

The **Certified Project Controls Professional — AI (PCP-AI)** is built for exactly this moment. It certifies the classical discipline of project controls **and** the judgement to use AI responsibly — under one clear principle: **AI proposes; the professional disposes**. The human stays accountable.

What it certifies

A PCP-AI holder can demonstrate capability across three areas:

- **Financial reporting & accounting (40%)** — the accounting model, IFRS/IAS-based reporting, revenue recognition on contracts, cost management and the commercial cycle. Controls professionals who understand the numbers behind the numbers.
- **Project management, controls & delivery (40%)** — budgeting and forecasting, performance and variance analysis, **earned value management**, scheduling and the critical path, contracts and commercial management, the project lifecycle, **agile and adaptive delivery**, risk, and process controls.
- **AI for project controls (20%)** — what AI genuinely is and is not, data quality, working with generative AI, the tool landscape, AI applied across the controls lifecycle, and — critically — the **governance, ethics and assurance** that keep a human accountable for every output.

Underneath sits the **PCP-AI Body of Knowledge**: thirteen domains, sixty-one Knowledge Areas, developed to full professional-handbook depth with worked examples, figures and practice questions.

Why it is different

- **AI is woven through, not bolted on.** Every domain shows where AI helps a specific task and where it must be governed — an honest treatment of real capabilities and real limits, not hype.
- **It is quantitative and applied.** Candidates compute earned-value indices, forecasts, variances, agile metrics and risk expected values — not just recall definitions.
- **It is examined securely and fairly.** A 90-minute, remotely-proctored examination, a 65% standard, immediate results with a domain-by-domain breakdown, and an interface available in seven languages.

Who should hold it

Cost engineers, planners and schedulers, project and commercial controllers, quantity surveyors, PMO analysts, and project-facing finance professionals who want a current, recognised credential that reflects both the discipline and its AI-augmented future.

The pathway

- 1. **Study** the Body of Knowledge and the published Content Outline.
- 2. **Register and pay** the examination fee — this opens a 12-month scheduling window.
- 3. **Sit** the 90-minute proctored examination, in the browser or the PCI Secure Exam application.
- 4. **Earn** a verifiable PCP-AI credential, and keep it current with continuing professional development on a three-year cycle.

At a glance

Credential	Certified Project Controls Professional — AI (PCP-AI)
Structure	13 domains · 61 Knowledge Areas · 3 competency areas (40% / 40% / 20%)
Examination	90 minutes · multiple choice · 65% to pass · remotely proctored
Languages	English, Arabic, Spanish, French, Korean, Chinese, Russian
Validity	Three-year cycle, maintained with CPD
Study resource	The PCP-AI Body of Knowledge

Project Controls Institute. Fees, dates and the final examination specification are confirmed on the official PCI website.

PCP-AI Sample Questions

Certified Project Controls Professional — AI (PCP-AI) · Project Controls Institute

Twelve sample questions illustrating the style, coverage and difficulty of the PCP-AI examination.

These are study items drawn from the same blueprint as, but kept separate from, the live examination bank; they are never used as live questions. The correct answer and a short rationale follow each item.

Area A — Financial Reporting & Accounting

1. [recall] A project entity reports total assets of \$4,200,000 and total liabilities of \$1,500,000. Applying the accounting equation, what is its equity? A. \$5,700,000 · **B. \$2,700,000** · C. \$1,500,000 · D. \$4,200,000 *Equity = Assets – Liabilities = 4,200,000 – 1,500,000 = \$2,700,000. Adding the two figures (A) is the common error.*

2. [recall] Under IFRS 15, revenue from a contract with a customer is recognised when (or as): **A. a performance obligation is satisfied** · B. cash is received · C. the invoice is raised · D. the financial year ends *IFRS 15 recognises revenue as control of the promised good or service transfers — i.e. as a performance obligation is satisfied — not on cash receipt or invoicing.*

3. [application] An asset costs \$50,000, has a residual value of \$5,000 and a useful life of 5 years. What is the annual straight-line depreciation charge? A. \$10,000 · B. \$11,000 · **C. \$9,000** · D. \$45,000 *(50,000 – 5,000) / 5 = 45,000 / 5 = \$9,000. Ignoring residual value gives \$10,000 (A).*

4. [analysis] Amounts billed to a customer on a contract exceed the revenue recognised to date. In the statement of financial position this gives rise to a: A. contract asset · **B. contract liability** · C. trade receivable · D. prepayment *Billing ahead of recognised revenue creates a contract liability (deferred revenue); recognising ahead of billing would create a contract asset.*

Area B — Project Management, Controls & Delivery

5. [application] Earned value (EV) is \$90,000 and actual cost (AC) is \$120,000. What is the cost performance index (CPI)? **A. 0.75** · B. 1.33 · C. 0.90 · D. 1.00 *CPI = EV / AC = 90,000 / 120,000 = 0.75 (under-performing on cost). Inverting the ratio gives 1.33 (B).*

6. [application] Budget at completion (BAC) is \$1,000,000 and the CPI to date is 0.80. Assuming past cost efficiency continues, what is the estimate at completion? A. \$800,000 · **B. \$1,250,000** · C. \$1,200,000 · D. \$1,000,000 *EAC = BAC / CPI = 1,000,000 / 0.80 = \$1,250,000.*

7. [application] Budget at completion is \$1,000,000, earned value is \$400,000 and actual cost is \$500,000. What is the to-complete performance index (TCPI) to finish on the original budget? A. 0.80 · **B. 1.20** · C. 1.00 · D. 0.83 *TCPI = (BAC – EV) / (BAC – AC) = 600,000 / 500,000 = 1.20 — the efficiency now required on remaining work.*

8. [analysis] An activity on a project network has a total float of zero. This activity is: A. non-critical · B. a milestone · C. a dummy activity · **D. on the critical path** *Zero total float means any delay to the activity delays the project — the definition of a critical-path activity.*

9. [application] A risk has a 20% probability of occurring and a \$50,000 impact if it does. What is its expected monetary value (EMV)? A. \$50,000 · B. \$250,000 · C. \$70,000 · **D. \$10,000** *EMV = probability × impact = 0.20 × 50,000 = \$10,000.*

10. [application] An agile team completes 24 story points across three sprints. What is its average velocity? A. 24 points/sprint · **B. 8 points/sprint** · C. 72 points/sprint · D. 3 points/sprint *Velocity = points completed / sprints = 24 / 3 = 8 points per sprint.*

Area C — AI for Project Controls

11. [recall] In project controls, the governing principle "AI proposes; the professional disposes" means that: **A. the professional remains accountable for the final decision** · B. AI outputs can be accepted as always correct · C. AI replaces the cost engineer · D. forecasts are issued automatically without review *The principle keeps a qualified human accountable for every estimate, forecast and decision, however the output was generated.*

12. [analysis] A generative-AI assistant drafts a cost narrative from a project's ledger, but the ledger is missing an accrual for work done and not yet invoiced. The most likely consequence is that the AI: A. corrects the missing accrual automatically · B. refuses to produce a narrative · **C. produces a fluent but wrong narrative, overstating cost performance** · D. flags the data as incomplete on its own *AI reasons over the data it is given. An incomplete input yields a confident, well-written, but incorrect output — which is exactly why the professional must verify the inputs and the result.*

Answer key

1-B · 2-A · 3-C · 4-B · 5-A · 6-B · 7-B · 8-D · 9-D · 10-B · 11-A · 12-C

Project Controls Institute. For the full treatment of every topic — with worked examples, figures and further practice — study the PCP-AI Body of Knowledge.